Tathagata Saha

Researcher · Astronomy

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Research Positions

Inter-University Centre for Astronomy and Astrophysics (IUCAA) *Pune, India*
Post Doctoral Fellow *15.09.2025 onwards*
as a member of the eROSITA German Consortium, with Prof. Chandreyee Maitra

Leibniz Institute for Astrophysics Potsdam (AIP) *Potsdam, Germany*
Research Assistantship and guest researcher *01.12.2023 - 31.05.2026*
with Dr. Mirko Krumpe

Nicolaus Copernicus Astronomical Center of the Polish Academy of Sciences *Warsaw, Poland*
PhD Astronomy *01.10.2019 - 22.03.2024*
Thesis Supervisor: Prof. dr. hab. Alex G. Markowitz

Education

Presidency University *Kolkata, India*
M.Sc. Physics *01.08.2017 - 01.07.2019*
- Thesis: Connection Between the Accretion Disc and Corona in the Active Galactic Nuclei
  Advisors: Prof. Ritaban Chatterjee, Prof. Indranil Chattopadhyay

Presidency University *Kolkata, India*
B.Sc.(Honours) Physics *21.07.2014-01.07.2017*
- Thesis: Accretion Disc Corona Interaction in Active Galactic Nuclei
  Honours thesis/undergraduate research advisor: Prof. Ritaban Chatterjee

Research Interests

- Accretion physics across the compact-object mass scale, from stellar-mass X-ray binaries to supermassive black holes in AGN.
- Compact object populations in the Magellanic Bridge, combining X-ray spectral and variability analysis with optical counterpart identification from Gaia to characterize accreting binaries and their connection to the underlying stellar population.
- Multi-wavelength spectral and timing studies of supermassive black hole transients, particularly changing-look AGN (CLAGN) and tidal disruption events (TDEs), to probe rapid accretion state changes.
- Time-resolved X-ray spectroscopy of radio-quiet Seyferts to constrain the structure and evolution of accretion sub-components, including the disk, corona, and broad-line region.
- Geometry and evolution of the circumnuclear obscuring medium (torus) and its connection to accretion inflows and outflows in AGN.
- Accretion-driven feedback and its role in black hole growth and galaxy evolution, including its imprint on the cosmic X-ray background and large-scale structure.
- Modeling of disk–corona interactions, jet formation, and the instabilities governing accretion state transitions.

Research Skills

- **Data analysis:** X-ray and optical data reduction and analysis using classical and Bayesian methods, including nested sampling (MultiNest, UltraNest) and Markov Chain Monte Carlo, within XSPEC, Sherpa, and

custom Python frameworks.

- **Observing proposals:** Contributions to proposals for optical (SALT, ESO-VLT) and X-ray (XMM-Newton, Chandra, NuSTAR, NICER, AstroSat) facilities, including feasibility simulations and development of the scientific case.
- **Large datasets:** Handling and cross-matching of large datasets from the eROSITA all-sky survey, Gaia, and the Legacy Survey using Python and SQL databases.
- **Programming:** Proficient in Python, with experience in numerical methods, data wrangling (NumPy, Pandas), statistical analysis (SciPy, scikit-learn), and deep learning (TensorFlow).
- **Academic service:** Referee for the journals of the American Astronomical Society (AAS), including The Astrophysical Journal (ApJ).

Publications

Refereed and accepted Publications

Saha. T et al., eRASSU J043115.8-711730: The first pulsating symbiotic super-soft X-ray source in the Magellanic Bridge, 2026, Accepted for publication in MNRAS; 2026

Saha. T et al., Multiwavelength study of extreme variability in LEDA 1154204: A changing-look event in a type 1.9 Seyfert, A&A, 702, A28; 2025

Saha. T et al., The Close AGN Reference Survey (CARS). Long-term spectral variability study of the changing look AGN Mrk 1018; A&A, 699, A205; 2025

Saha, T; Markowitz, A. G.; Buchner, J, Inferring the morphology of AGN torus using X-ray spectra: a reliability study; MNRAS, 509, 5485; 2022

Mondal, A; **Saha, T**; Sar, A; and Chatterjee, R.; Dynamic Accretion Disk–Corona Connection and Broad Fe Kα Variability in Mrk 766 Revealed by Time-resolved High-resolution X-Ray Spectroscopic Analysis; ApJ, 1011, 31

Palit, B. ; Rozanska, A. ; Markowitz, A. G. ; Lawther, D. ; Vestergaard, M. ; Ruan, J. J. ; **Saha, T.** et al. ; A break in the X-ray loudness of Markarian 590: evidence for an AGN spectral state transition? Accepted for Publication in ApJS: arXiv:2602.17572

Homan, D., Krumpe M., Markowitz, A., **Saha, T.**, and 16 co-authors. Discovery of the luminous X-ray ignition eRASSt J234402.9-352640; I. Tidal disruption event or a rapid increase in accretion in an active galactic nucleus?, A&A 672, A167; 2023

Laha, S., Markowitz, A., Krumpe, M., Nikutta, R., Rothschild, R., **Saha, T.**. The Variable and Non-variable X-Ray Absorbers in Compton-thin Type II Active Galactic Nuclei, ApJ, 897, 66; 2020

Markowitz A., Krumpe M., ... **Saha, T.** and 13 co-authors. eROSITA Detection of a Cloud Obscuration Event in the Seyfert AGN EC 04570-5206, 2024, A&A, 684, A101; 2024

Krishnan, S., Markowitz, A., ... **Saha, T.** and 13 co-authors. An X-ray flaring event and a variable soft X-ray excess in the Seyfert LCRS B040659.9-385922 as detected with eROSITA, A&A, 691, 19; 2024

Markowitz A., Krumpe M., ... **Saha, T.** and 12 co-authors. A changing-look Seyfert discovered by eROSITA reveals a two-component broad-line region, A&A, 710, A222; 2026

Mereghetti, S et al.....**Saha. T**, Time domain astronomy with the THESEUS satellite, Exp Astron, 52, 309; 2021

Ongoing Projects

- Characterizing the stellar and compact object populations in the Magellanic Bridge using eROSITA detections, GAIA identifications and multi-wavelength follow-up using *XMM-Newton*.
- Development of a neural network based spectral classifier that can facilitate identification of changing look AGNs by flagging spectral type changes in optical spectral surveys (e.g. SDSS).
  Github repository: `github.com/tathagatas1996/AGNoptspecNN-base`
- Development of a code base (Saha & Naddaf in prep.) to study the structural evolving of the broad line region in transient changing-look AGNs using Failed Radiatively Driven Outflow model (Naddaf et al., 2022, A&A, 663, A77)

Teaching Experience

August 2024 — **X-ray Data Analysis with XSPEC**, **Content:** X-ray data and statistic, Likelihood functions, Bayesian Statistics: Markov Chain Monte Carlo (MCMC), Nested Sampling, PyXSPEC , **Location:** School of Astrophysics, Presidency University, Kolkata

Scholarships & Grants

2023 — **Research Assistant Scholarship**, Deutsches Zentrum für Luft und Raumfahrt (DLR), P.I. Dr. Mirko Krumpe, FKZ 50 OR 2004

2020 — **Ph.D. Scholarship on Active Galactic Nuclei**, Polish National Science Center (NCN), P.I. Prof. dr. hab. Alex G. Markowitz, 2018/31/G/ST9/03224

2017 — **Pre-Ph.D. Selection with Scholarship**, Inter-University Center for Astronomy and Astrophysics (IUCAA), Pune, India

2017 — **Integrated Ph.D. Selection with Scholarship**, Department of Physics, Indian Institute of Science, Bengaluru, India

2017 — **All India Rank 26, Joint Entrance Screening Test for Ph.D. programs** ,

Presentations

Invited Oral Presentations

January 2022, *A BXA-driven study of reliability of X-ray Spectral fits in determining AGN torus morphology*. Invited talk: X-ray Spectral Fitting 2022 (XSF22), Prague, Czech Republic

April 2023, *Active Galactic Nuclei: Active Galactic Nuclei: The multi-wavelength signature of black hole accretion*. Invited Outreach talk: Kolkata, India.

September 2024, *Morphology of circumnuclear accreting gas in Active Galactic Nuclei*, Invited Talk, School of Astrophysics, Presidency University, Kolkata

April 2025, *Multiwavelength study of the accretion substructures in AGN transients*, Invited Talk, CESSI, IISER, Kolkata

July 2025, *Multiwavelength study of the accretion substructures in AGN transients*, Invited Talk, High Energy Astrophysics Group, IUCAA, Pune

Contributed Oral Presentations

September 2021, *Determining the AGN torus morphology using X-ray Spectra: A reliability study*, Polish Astronoical Society Meeting 2021, Szczecin, Poland

July 2022, *A BXA-driven study of reliability of X-ray Spectral fits in determining AGN torus morphology*, Sexten Center for Astrophysics, Sexten, Italy

December 2022, *A BXA-driven study of reliability of X-ray Spectral fits in determining AGN torus morphology*, TORUS2022 Conference, Leiden, Netherlands

March 2023, *Multiwavelength follow-up studies of eROSITA detected AGN transients*, Astronomical Society of India Meeting 2023, Indian Institute of Technology, Indore, India

June 2023, *Multiwavelength follow-up study of a flaring changing-look AGN with double-peaked Balmer lines,* SALT Conference 2023, Warsaw

October 2023, *Multiwavelength follow-up study of a flaring changing-look AGN with double-peaked Balmer lines* Gaia Alerts Conference 2023, Malta

July 2024, *Multiwavelength follow-up studies of eROSITA detected AGN transients*, Conference on Blazars and the Restless AGN, Presidency University, Kolkata, India

June 2026, *J0431-71: The first symbiotic supersoft source in the Magellanic Bridge*, X-ray Universe, Elche, Spain

July 2026, *Changing look AGNs: divergence in the observed timescales of state transition*, Conference on Blazars and the Restless AGN, IISER, Mohali, India

Contributed Poster Presentation

July 2021, *Determination of AGN Torus Mophology with X-ray Spectra: A reliability study*, European Astronomical Society Meeting 2021

March 2022, *Determination of AGN Torus Mophology with X-ray Spectra: A reliability study*, Astronomical Society of India Meeting 2023, Aryabhatta Research Institute for Observational Sciences, Nainital, India

July 2022, *Determination of AGN Torus Mophology with X-ray Spectra: A reliability study*, European Astronomical Society Meeting 2022

May 2022, *Multi-wavelength study of extragalactic transients detected with eROSITA: A flaring event in an AGN*, XMM-Newton Worshop 2022, Villafranca del Castillo, Madrid, Spain

June 2023, *Multi-wavelength study of a flaring CLAGN with double-peaked Balmer lines detected with eROSITA and ZTF*, The Restless AGN, Napoli, Italy

July 2023, *Multi-wavelength study of a flaring CLAGN with double-peaked Balmer lines detected with eROSITA and ZTF*, European Astronomical Society Meeting 2023, Krakow

Mentoring

2026	**Mahek Patel**, Florida Institute of Technology, USA	*IUCAA*
2026	**Ayon Mondal**, University of Maryland, Baltimore County, USA	*AIP, IUCAA*
2026	**Fida Fathima SS**, Presidency University, offered Ph.D. Position at University of Atacama	*IUCAA*

Outreach, Service, & Professional Development

January 2027	**Intl. Conference: X-ray Polarization of Compact Objects**, member of the Young Scientific Organizing Committee (YSOC)	*IUCAA, Pune, India*
June 2023	**Intl. Conference: SALT 2023 Conference, Warsaw**, member of the Local Organizing Committee	*CAMK-PAN, Warsaw, Poland*
2021-	**Theseus Mission**, Contributing Scientist, changing look AGNs with the Theseus Mission	
2023-	**Astronova Club, IEEE, Calcutta University**, Membership	*Kolkata, India*